6. MASS AND BALANCE

6.1 LOADING DATA

When calculating the CG of an external load use the arm of the hook assembly, i.e. 3300 mm (129.9 in).

6.2 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
External Cargo Hook System	D134-387	12.4	3300	40920
External Cargo Hook System	105-80322	12.4	3300	40920
– Installation Provisions	105-80344	4.9	3300	16170
– Detachable Items	105-80332	7.5	3300	24750
External Cargo Hook System	105-80325	11.6	3300	38280
– Installation Provisions	105-80345	5.1	3300	16830
– Detachable Items	105-80337	6.5	3300	21450
External Cargo Hook System	105-803411	14.58	3095	45125
– Installation Provisions	105-803471	6.01	3027	18192
– Detachable Items	117-80127	8.57	3143	26935
Hook Retraction Equipment	105-80501	1.91	3076	5857
– Installation Provisions	105-80503	0.09	557	50
– Detachable Items	105-80506	1.82	3200	5824

7. SYSTEM DESCRIPTION

The external cargo hook system allows for utilization of the helicopter for transporting external cargo loads having a mass of up to max 900 kg (1984 lb). The external cargo hook system comprises a hook assembly, steel suspension cables, clamps secured to the landing gear cross tubes, and electrical and mechanical hook releases.

Three hook assemblies are available as detachable items, P/N 105-80332, 105-80337 and 117-80127. These hook assemblies are not always interchangeable due to differing helicopter electrical systems and hook release circuitries. Refer to paragraph 8 this supplement for the combinations of hook assemblies, fixed provisions, and helicopter serial numbers that are compatible.

EFFECTIVITY *Hook Assembly P/N 105-80332*

The hook assembly (see Fig.5) is equipped with a rear-mounted grip having a trigger. This trigger is for use by the ground crew member to close the hook and thereby arm the pilot-activated opening mechanism. A cable loop is fastened at one end to the hook assembly.

To attach a load to the hook, the free-end of the cable loop is passed through the external load sling or attach ring and held in the hook assembly opening. Pulling the trigger then releases a locking claw to secure the loop free-end to the hook assembly.

A retainer mounted to the lower LH fuselage provides for securing the hook assembly when not in use. When the retainer is of the snap-hook type, the cable loop is used to secure the hook assembly in its stowed position. However, when the retainer is of the safety-rope type, then the hook assembly is positioned so that its locking claw secures the safety rope free-end. If the hook assembly is properly stowed the pilot can release it while in flight by activating either the electrical or mechanical releases.

Placing the LOAD HOOK switch on the collective lever in the OPEN position supplies 28V DC to open the hook assembly locking claw and cause the **HOOK** caution light on the switch panel to come on. The LOAD HOOK switch must then be placed in the CLOSED position to allow for manual closing of the locking claw. The hook assembly, however, remains open until being closed by pulling the trigger on the hook assembly. An open hook assembly is indicated by the **HOOK** caution light coming on.

EFFECTIVITY *Hook Assemblies P/N 105-80337 and 117-80127*

The hook assembly (see Fig.5) is installed with the opening of the hook facing either aft (P/N 105-80337) or forward (P/N 117-80127). A spring-loaded catch closes off the opening when the hook is latched closed. When the pilot activates the hook release (either electrical or mechanical), the hook pivots downwards allowing the load (minimum 5 kg to overcome spring pressure) to fall free. A spring then returns the hook to its closed position. The hook can also be released by ground crewmembers pulling a manual release knob provided on the hook assembly.

A retainer mounted to the lower LH fuselage provides for securing the hook assembly when not in use. The pilot can release the hook assembly from its stowed position while in flight by activating either the electrical or mechanical releases. An unloaded, trailing hook assembly is sensed by a microswitch within the hook assembly unit and indicated by the **HOOK** caution light coming on regardless of the LOAD HOOK switch position.

Placing the LOAD HOOK switch on the collective lever in the OPEN position supplies 28V DC to energize the hook release circuit. The **HOOK** caution light comes on to indicate that the load is released and that the hook assembly is in its operating position (not stowed). The LOAD HOOK switch must be placed in the CLOSED position to latch the hook closed.

EFFECTIVITY *Hook Retraction Equipment, if installed*

The hook assembly retainer, mounted to the lower LH fuselage, is replaced by a hook retraction winch. The winch cable is secured to the cargo hook assembly to allow for retraction and extension of the hook during flight.

The electrically-driven winch is supplied 28V DC via the HOOK WINCH circuit breaker and operated by the HOOK WINCH switch both located on the switch panel. Winch operation is indicated by illumination of a MOVE advisory light also located on the switch panel.

Placing the HOOK WINCH switch in the EXTEND position powers the winch motor in a direction to lower the cargo hook assembly. When the hook reaches its operating position, indicated by the EXTD'D advisory light coming on, limit switches within the winch motor stop the winch and cause the MOVE advisory light to go off. Placing the HOOK WINCH switch in the RETRACT position will then cause the winch motor to operate in the opposite direction to retract the hook assembly. The hook is fully retracted when both the EXTD'D and MOVE advisory lights go off.

EFFECTIVITY *ALL*

A mechanical release kick lever mounted on the floor aft of the pilot's directional control pedals is used for releasing the load in an emergency.

An external rear-view mirror P/N 105-80441 (D131-1231 or D131-79) can be installed to aid the pilot during load pick up and release.

NOTE The mirror shall be either covered or removed for night operations to avoid blinding effects when using the landing light.

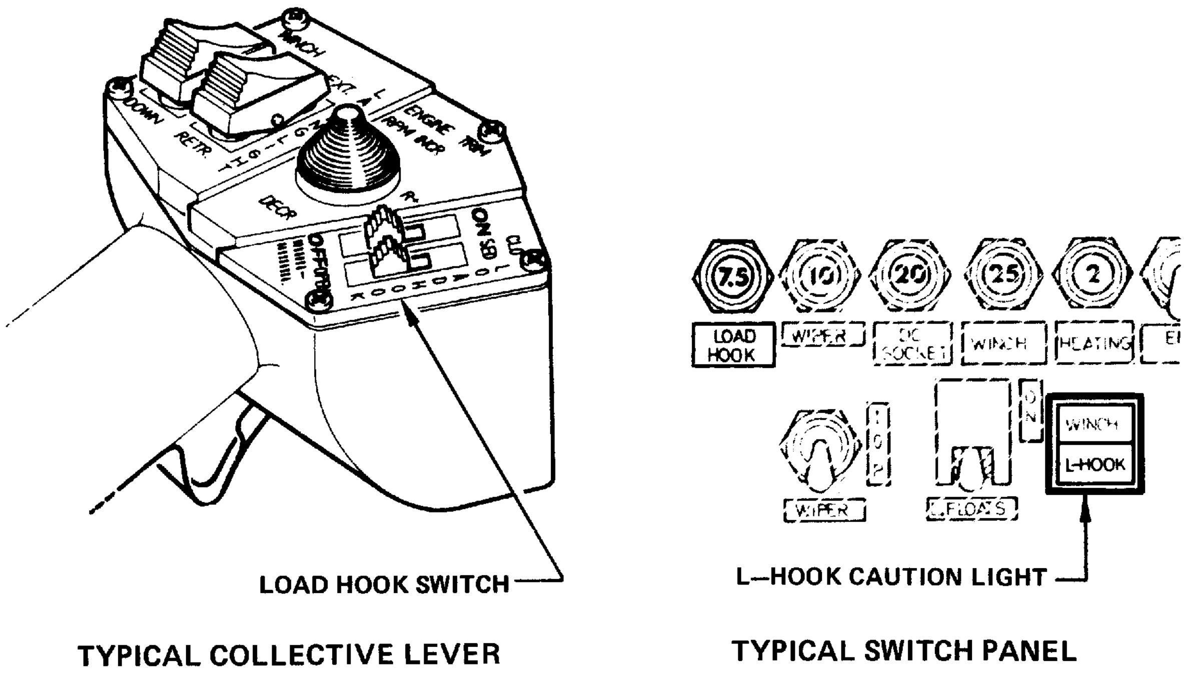

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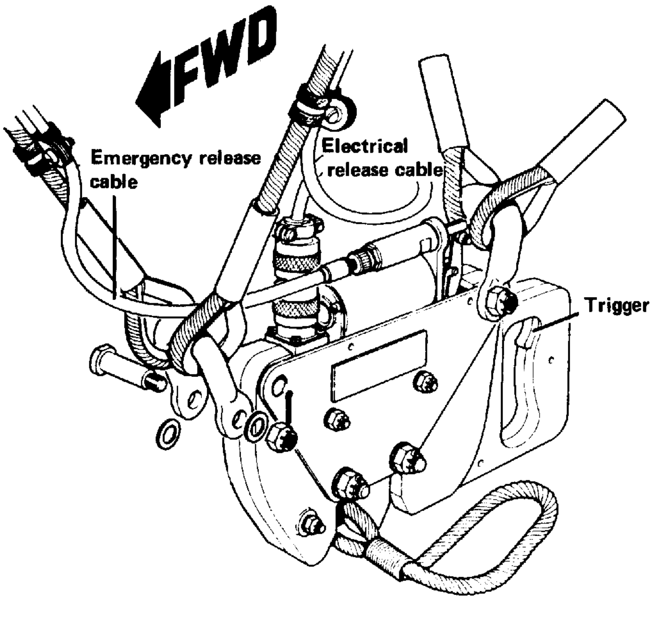
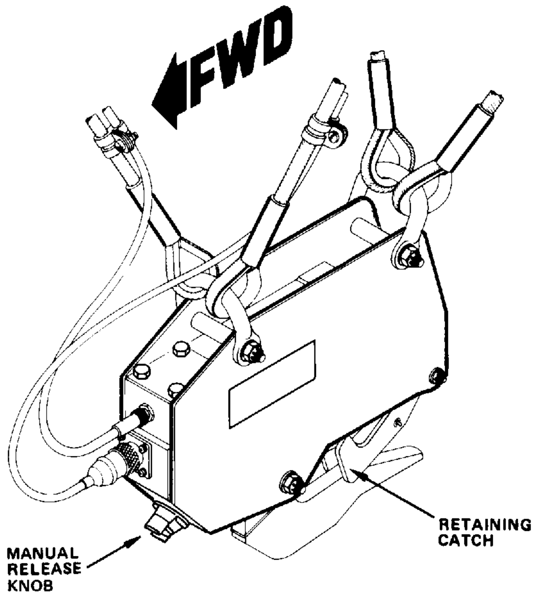
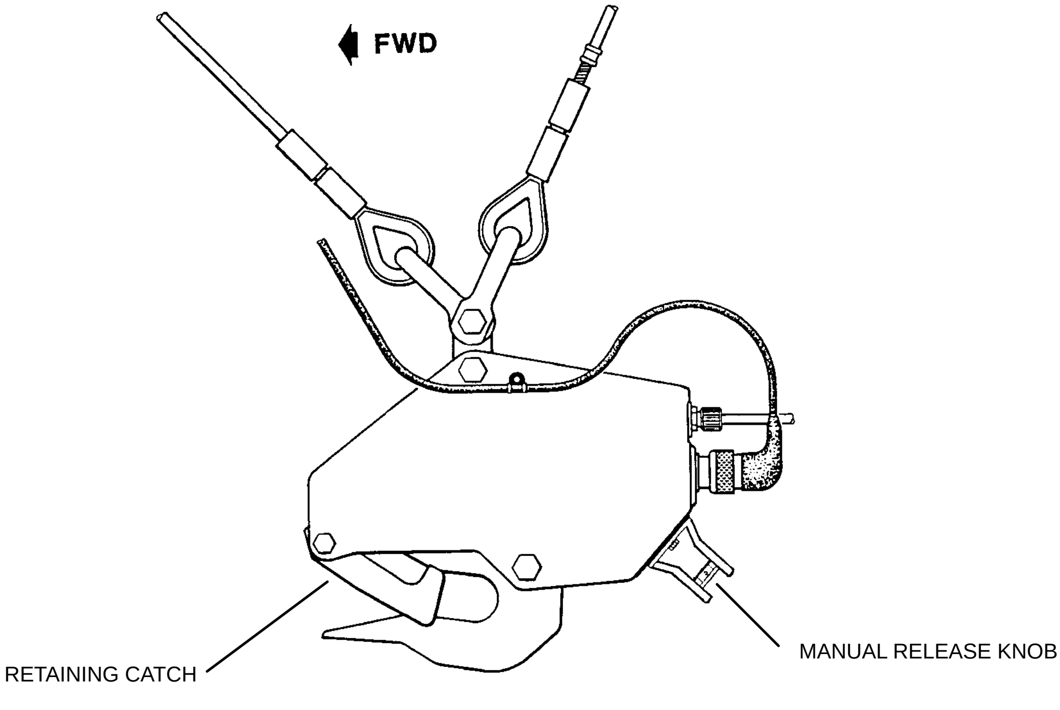
Fig.5 External Cargo Hook System Components (Sheet 1 of 2)



HOOK ASSEMBLY, P/N 105-80332



HOOK ASSEMBLY, P/N 105-80337



HOOK ASSEMBLY, P/N 117-80127

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Fig.6 External Cargo Hook System Components (Sheet 2 of 2)